Expected Goals (xG) Modelling in Football

Literature Review and Project Plan

ACM 40960 · Mathematical Modelling

Background

Expected Goals (xG) is a way of measuring how likely a shot is to end up as a goal. Each shot is given a value between 0 and 1, where a higher value means a better chance of scoring, based on features such as the distance from goal, the angle to goal, the body part used, and the amount of defensive pressure. Over the last decade xG has become a standard metric in football analytics and is now used widely by clubs, broadcasters, and researchers. Because goals are rare, a measure that estimates the quality of a shot rather than just counting goals gives a more stable picture of performance, which is part of why the metric has become so popular.

Literature Review

Early xG models were kept deliberately simple. Rathke (2017) showed that a logistic regression model based mainly on shot distance and angle could already give a reasonable estimate of goal probability. This set the basic template that most later work has built on.

Much of the later research has focused on adding more information and using more flexible methods. Lucey et al. (2015) showed that strategic and spatial features, such as the positions of nearby defenders, improved prediction beyond shot location alone. Anzer and Bauer (2021) extended this by combining synchronised positional and event data to build one of the more detailed models in the field. Mead et al. (2023) argued that standard models leave out useful information such as player and team ability, and showed that adding these features changed the model outputs in a meaningful way. Alongside richer features, researchers have moved from logistic regression toward tree based ensemble methods such as Random Forest and gradient boosting, which handle non-linear patterns more easily. Bandara et al. (2024) went a step further by using the sequence of events before a shot, suggesting that the build-up play carries information that static shot features miss.

A separate strand of work has focused on interpretability and on adjusting models for who takes the shot. Hewitt and Karakuş (2023) trained position adjusted models for forwards, midfielders, and defenders, and found clear differences between them. Scholtes and Karakuş (2024) used a Bayesian hierarchical approach, called Bayes-xG, to correct for player and position while sharing information across players. Iapteff et al. (2025) used Bayesian mixed models with a similar aim of keeping the model interpretable rather than treating it as a black box. These studies reflect a growing interest in models that are not only accurate but also explainable, which matters if coaches and analysts are going to trust the output.

A common use of xG is to judge finishing ability by comparing a player's actual goals to their total xG, often called goals above expectation. This use has been questioned. Davis and Robberechts (2024) found that a player's overperformance is not stable from one season to the next, and that xG models can carry biases because stronger players and teams take more shots, which skews the data the model learns from. Their work is a useful reminder that a finishing table based on actual goals minus xG should be read with care rather than treated as a clean measure of skill.

Across these studies, models are usually judged using AUC-ROC, the Brier score, and calibration plots. AUC measures how well a model separates goals from non-goals, while the Brier score and calibration plots check whether the predicted probabilities are realistic, for example whether shots given an xG of about 0.3 really are scored around 30 percent of the time. Calibration is often treated as just as important as raw accuracy, since an xG model is only useful if its probabilities can be trusted.

xG is also a natural building block for modelling whole match outcomes. There is a long line of work that models the number of goals a team scores using Poisson based models, starting with Dixon and Coles (1997), who used a Poisson regression with team attack and defence strengths to model scores and study the betting market. Baio and Blangiardo (2010) later placed this kind of model in a Bayesian hierarchical framework. Because each shot already has a goal probability, the goals in a match can be treated as the sum of independent Bernoulli trials, which leads to a Poisson-binomial distribution and allows match outcomes and expected points to be simulated directly from the shots. This links the shot level xG model to a simulation of results at the match level.

Overall, the literature has moved from simple distance and angle models toward richer, more flexible, and more interpretable ones. However, many of the more advanced models depend on detailed tracking data that is not always openly available, and a lot of comparisons focus on accuracy while paying less attention to calibration, to the known limitations of finishing measures, and to the mathematical structure behind the model. This project treats xG as a mathematical model rather than only a prediction task. The shot geometry is derived from first principles, the fitting of the model is framed as a maximum likelihood optimisation problem, the model is extended to match outcomes through Monte Carlo simulation, and the uncertainty in the predictions is quantified. The work is carried out on the StatsBomb open dataset so that it stays reproducible.

Project Plan

- **Data setup:** Shot events will be extracted from the StatsBomb open dataset, including the shot coordinates, the body part used, the assist type, the pressure, and the technique.

- **Exploratory data analysis:** Shot maps and conversion rates will be visualised. The data will also be checked for class imbalance, and the categorical features will be encoded so they can be used by the models.

- **Shot geometry:** The distance and angle features will be derived from the pitch coordinates using basic trigonometry, where the angle is the width of the goal as seen from the shot location. Iso-distance and iso-angle curves will be drawn on the pitch to show how goal probability depends on position.

- **Baseline model:** A logistic regression model will be fitted using shot distance and angle. The model will be written out in full as a log-odds model, and the fitting will be presented as a maximum likelihood optimisation problem. It will then be evaluated using AUC, the Brier score, and a calibration plot.

- **Enhanced models:** Additional features will then be added, and a Random Forest and an XGBoost model will be compared against the baseline. Calibration will be assessed for each model, not just ranking accuracy. Player finishing ability will be explored by comparing actual goals to xG, while keeping in mind the known biases in this measure.

- **Bayesian model and uncertainty:** A Bayesian version of the model will be fitted in Stan so that each xG value comes with a credible interval rather than a single number. This quantifies how confident the model is and feeds into the calibration and finishing discussion.

- **Match simulation:** Using the fitted xG values, the goals in a match will be modelled as a sum of Bernoulli trials, giving a Poisson-binomial distribution. A Monte Carlo simulation will then estimate win, draw, and loss probabilities and the expected points a team deserved from its shots.

- **Sensitivity analysis:** A sensitivity analysis will show how the predicted goal probability responds to changes in distance and angle, so the behaviour of the model is made clear rather than left as a black box.

- **Write-up:** Finally, a full report will be produced, covering the derivations, the model comparisons, the simulation results, the visualisations, and a discussion of the limitations.

Data

The main data source for this project is the StatsBomb open dataset, which is available at the link given in the references. It contains shot level event data from competitions such as La Liga, the FA Women's Super League, and UEFA EURO 2020. The data can be accessed through the `statsbombpy` Python library.

Expected Outcomes

The project will produce a calibrated xG model that compares logistic regression, Random Forest, and XGBoost, together with a derivation of the shot geometry and a treatment of the model as a maximum likelihood optimisation problem. A Bayesian version will provide uncertainty estimates for the predictions. Building on the fitted model, a Monte Carlo simulation will produce match outcome probabilities and expected points. The project will also produce pitch heatmaps, calibration plots, a sensitivity analysis, and a player finishing table, with the finishing table interpreted with appropriate caution given the known limitations of this measure.

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